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Demonstrating the Ratio of Time Dilation Between Two Systems Emerges Solely From Their Relative Respective Mass to Radius Ratios

Thomas Damon DeGerlia[a]

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ABSTRACT: All measurement is intrinsically comparative: no measured quantity has physical meaning without a comparator. This paper demonstrates a direct relationship between the time dilation between two systems and their respective spatial mass distribution by calculating and verifying the relative pace of time between uniform spherical systems from comparative geometry alone. Using the compactness relation $D = M/R$ (DeGerlia compactness), the governing parameter is the ratio of mass-over-radius terms, $D_1/D_2 = M_1R_2/M_2R_1$, with the Schwarzschild collapse condition expressed as the universal threshold $D_{\text{crit}} = c^2/2G = 6.73295 \times 10^{26}$ kg/m. Time dilation then follows from the same form in every case, $\sqrt{1-k}$, where k is the relevant compactness-derived geometric ratio. Across static mass, static radius, constant density, constant inertia, and fully unconstrained mass-radius comparisons, the compactness-ratio calculation returns the Schwarzschild time-dilation result exactly. This establishes that the relative rate of time between systems is not an ad hoc relativistic output requiring separate formulas, but a direct consequence of comparative mass-radius geometry. The result makes explicit a simple relation clearly contained in general relativity: given two systems' masses and mean radii, their relative time dilation is determined by their respective geometric distribution of that mass.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia compactness, inertial density, moment of inertia, Schwarzschild radius

1 Introduction

General relativity is a model for observing our universe that is fundamentally comparative: no quantity has physical meaning without a comparator. The pace of time is no exception: the time dilation between two systems is a a product of the systems respective geometric mass distribution. Directly from the gravittional time dilation formula one can derive the underlying two-system logic and ultimately GTD is simply a comparison between a system a systems pace of time relative to the pace of time of an arbitrarily selected benchmark system, which in this case is the same system at its Schwarzschild radius. The formula allows you the freedom to only insert one mass and one radius, and the second is derived directly from the first. This has the effect of hiding the true nature of relativity and that is that it is, above all, comparative. But sends a signal that one need not compare two systems to understand time, and that is flawed thinking; it introduces the mathematical fallacy of an arbitrarily selected comparator, when there is, in fact, no privilege perspective from which to understand time. In this paper we evaluate the physical property responsible for relativistic time dilation and derive the underlying formulation that quantifies the relative time dilation between two systems.

The relative flow of time between two systems is dictated by their relative compactness. The standard formula for compactness, described under general relativity, expresses this as a dimensionless parameter. However, in this paper we utilize a principle developed by the author, DeGerlia Compactness, which is an abstracted compactness measure explained herein, that retains the proper unit of this property and that is mass over radius. Thus DeGerlia compactness is defined as a system's mass over mean radius $D = m/\bar{r}$[1]. We do a comparative analysis and tabulate relative time dilation across a variety of systems from classical to relativistic regimes, and show that the same compactness ratio $k = D_1/D_2 = M_1R_2/(M_2R_1)$ returns the Schwarzschild time-dilation ratio exactly across every tested constraint class, including the fully general case. This establishes that the relative

[a] DeGerlia Expert Consulting, 3000 Lawrence Street, Denver, CO, United States of America. E-mail: tom.degerlia@tomdegerlia.com

rate of time between systems is not an ad hoc relativistic output requiring separate formulas, but a direct consequence of comparative mass-radius geometry. The result makes explicit a simple relation already contained in general relativity: given two systems' masses and mean radii, their relative time dilation is determined by their respective geometric distribution of mass alone.

2 General Relativity and Compactness

2.1 Compactness

Compactness is the dimensionless ratio of a system's mass to its radius, scaled by G/c^2. The most common forms are $C = GM/(Rc^2)$ and $r_s/R = 2GM/(Rc^2) = 2C$, where $r_s = 2GM/c^2$ is the Schwarzschild radius. Both measure how close a system sits to its Schwarzschild threshold: $r_s/R = 1$ marks the gravitational-collapse condition.

2.2 Inertial Density P

Inertial density P[1] is the moment of inertia of a system divided by its volume:

$$P = I/V \tag{1}$$

Inertia, in a linear context, is simply mass. Inertial density, in this context, reduces directly to mass density $\rho = M/V$. From this perspective, inertial density is simply the rotational analog to mass density, not unlike the relationship between f equals m a and torque equals angular acceleration times m. More succinctly stated: mass density is simply the linear interpretation of inertial density.

Substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, with R the radius, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \tag{2}$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

2.3 DeGerlia Compactness D

$D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system. Given that this quantity is almost always used in a ratio that would cancel out all geometric vectors and constants, for ease of use, this metric D for inertial density is typically used over the geometrically accurate inertial density, P.

2.4 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia compactness D. There are several advantages to this approach, not the least of which is that you don't have to calculate the threshold for every mass. And, it's much easier to calculate D than P. Inertial density is an intuitive metric that's easier to work with.

Setting $R = r_s$ (the Schwarzschild condition)[2] and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia compactness at which the escape velocity equals the speed of light:

$$D_{crit} = \frac{c^2}{2G} = (6.73295 \pm 0.00015) \times 10^{26}\ \mathrm{kg/m} \tag{3}$$

Note this value is exact; its precision is limited by $G = (6.67430 \pm 0.00015) \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$[3].

3 The Space-Time Equivalence

The following identity relates the time-dilation ratio of two uniform spherical systems to their mass and radius.*

$$k = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \tag{4}$$

where I_1, I_2 are the moments of inertia, R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia Compactness of system i. The three forms in Equation 4 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 exposes the compactness-ratio structure, and $M_1R_2/(M_2R_1)$ is the fully expanded form.

A system's pace of time has meaning only relative to another pace of time. The scale factor k governs the relative time-dilation between two uniform spherical systems. The standard gravitational time-dilation calculation takes the Schwarzschild form $\sqrt{1-k}$, where $k = D/D_{crit} = r_s/R$. The compactness ratio $k = D_1/D_2$ and the Schwarzschild form give the same value.

Via the Schwarzschild radius, the same scale factor is:

$$k = \frac{D}{D_{crit}} = \frac{r_s}{R} \tag{5}$$

where $D = M/R$ is the system's DeGerlia Compactness and $r_s = 2GM/c^2$ is its Schwarzschild radius. This is the k that appears in the standard $\sqrt{1-k}$ for GTD.

3.1 Relating Linear Scale Factor k to General Relativity

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{6}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the DeGerlia compactness ratio for the case being computed. The specific forms are:

- $k = D_1/D_2$ (Equation 4) — via inertial density ratio
- $k = D/D_{crit} = r_s/R$ (Equation 5) — via Schwarzschild radius

Same k symbol across forms: a given numerical k produces the same time-scale-factor result regardless of which form it appears in.

4 Relative Time Dilation Examples

Example parameter sets match the columns of the pair comparison tables in Appendix A.

* This identity has been previously referred to as the "space-time equivalence"[4].

4.1 Static Mass ($M_1 = M_2$)

When the two systems share the same mass, the mass terms cancel in k:

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{R_2}{R_1}$$

The compactness ratio reduces to a ratio of radii. Equivalently in terms of DeGerlia compactness $D = M/R$, this is $D_1/D_2 = R_2/R_1$; each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit} = r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

Example (Static Mass, benchmark): System 1 is the system being measured; System 2 is a same-mass reference at its Schwarzschild radius. The benchmark configuration sets $D_{norm,2} = 1$, the condition under which the two-system compactness ratio reduces to the standard one-system Schwarzschild GTD expression.

System properties:

S_1: $M_1 = 2.00000 \times 10^{30}$ kg, $R_1 = 3.00000 \times 10^3$ m, $\rho_1 = 1.76839 \times 10^{19}$ kg/m^3, $I_1 = 7.20000 \times 10^{36}$ kg·m^2.

S_2: $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3$ m, $\rho_2 = 1.82166 \times 10^{19}$ kg/m^3, $I_2 = 7.05893 \times 10^{36}$ kg·m^2.

DeGerlia compactness:

$$D_1 = \frac{M_1}{R_1} = \frac{2.00000 \times 10^{30}}{3.00000 \times 10^3} = 6.66667 \times 10^{26} \text{ kg/m}$$

$$D_2 = \frac{M_2}{R_2} = \frac{2.00000 \times 10^{30}}{2.97046 \times 10^3} = 6.73295 \times 10^{26} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = \frac{D_1}{D_{crit}} = 9.90155 \times 10^{-1}$$

$$D_{norm,2} = \frac{D_2}{D_{crit}} = 1.00000$$

Schwarzschild radii:

$$r_{s,1} = 2GM_1/c^2 = 2.97046 \times 10^3 \text{ m}$$

$$r_{s,2} = 2GM_2/c^2 = 2.97046 \times 10^3 \text{ m}$$

With $D_{norm,2} = 1$, the two-system compactness ratio reduces to System 1's individual compactness. Computed two ways:

$$k_s = r_{s,1}/R_1 = 9.90155 \times 10^{-1}$$

$$k_d = D_{norm,1}/D_{norm,2} = D_{norm,1} = 9.90155 \times 10^{-1}$$

GTD of System 1 against the benchmark:

$$\text{GTD}_s = \sqrt{1 - k_s} = \sqrt{1 - 9.90155 \times 10^{-1}} = 1 - 9.00777 \times 10^{-1}$$

$$\text{GTD}_d = \sqrt{1 - k_d} = \sqrt{1 - 9.90155 \times 10^{-1}} = 1 - 9.00780 \times 10^{-1}$$

Example (Static Mass, Table Case 1): System 1 is the system being measured; System 2 is a same-mass reference system at a different radius.

System properties:

S_1: $M_1 = 2.00000 \times 10^{30}$ kg, $R_1 = 3.00000 \times 10^3$ m, $\rho_1 = 1.76839 \times 10^{19}$ kg/m^3, $I_1 = 7.20000 \times 10^{36}$ kg·m^2.

S_2: $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = 7.00000 \times 10^8$ m, $\rho_2 = 1.39203 \times 10^3$ kg/m^3, $I_2 = 3.92000 \times 10^{47}$ kg·m^2.

DeGerlia compactness:

$$D_1 = \frac{M_1}{R_1} = \frac{2.00000 \times 10^{30}}{3.00000 \times 10^3} = 6.66667 \times 10^{26} \text{ kg/m}$$

$$D_2 = \frac{M_2}{R_2} = \frac{2.00000 \times 10^{30}}{7.00000 \times 10^8} = 2.85714 \times 10^{21} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = \frac{D_1}{D_{crit}} = 9.90155 \times 10^{-1}$$

$$D_{norm,2} = \frac{D_2}{D_{crit}} = 4.24352 \times 10^{-6}$$

Schwarzschild radii:

$$r_{s,1} = 2GM_1/c^2 = 2.97046 \times 10^3 \text{ m}$$

$$r_{s,2} = 2GM_2/c^2 = 2.97046 \times 10^3 \text{ m}$$

Compactness ratio for System 1, computed two ways:

$$k_s = r_{s,1}/R_1 = 9.90155 \times 10^{-1}$$

$$k_d = D_{norm,1} = 9.90155 \times 10^{-1}$$

GTD via r_s/R:

$$\text{GTD}_{s,1} = \sqrt{1 - r_{s,1}/R_1} = 1 - 9.00777 \times 10^{-1}$$

$$\text{GTD}_{s,2} = \sqrt{1 - r_{s,2}/R_2} = 1 - 2.12176 \times 10^{-6}$$

GTD via D/D_{crit}:

$$\text{GTD}_{d,1} = \sqrt{1 - D_{norm,1}} = 1 - 9.00780 \times 10^{-1}$$

$$\text{GTD}_{d,2} = \sqrt{1 - D_{norm,2}} = 1 - 2.12176 \times 10^{-6}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\text{GTD}_{s,1}/\text{GTD}_{s,2} = \frac{1 - 9.00777 \times 10^{-1}}{1 - 2.12176 \times 10^{-6}} = 1 - 9.00776 \times 10^{-1}$$

$$\text{GTD}_{d,1}/\text{GTD}_{d,2} = \frac{1 - 9.00780 \times 10^{-1}}{1 - 2.12176 \times 10^{-6}} = 1 - 9.00780 \times 10^{-1}$$

All values above appear in column $M = M$ of Table 1 in Appendix A.

4.2 Static Radius ($R_1 = R_2$)

When the two systems share the same radius, the radius terms cancel in k:

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{M_1}{M_2}$$

The compactness ratio reduces to a ratio of masses. The Lorentz factor (Equation **??**) has the same form as Equation 6, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling.

Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

Example (Static Radius, Table Case 2): System 1 is the system being measured; System 2 is an equal-radius reference system at a different mass.

System properties:

S_1: $M_1 = 1.00000 \times 10^{30}$ kg, $R_1 = 1.00000 \times 10^8$ m, $\rho_1 = 2.38732 \times 10^5$ kg/m^3, $I_1 = 4.00000 \times 10^{45}$ kg·m^2.

S_2: $M_2 = 1.00000 \times 10^{24}$ kg, $R_2 = 1.00000 \times 10^8$ m, $\rho_2 = 2.38732 \times 10^{-1}$ kg/m^3, $I_2 = 4.00000 \times 10^{39}$ kg·m^2.

DeGerlia compactness:

$$D_1 = \frac{M_1}{R_1} = \frac{1.00000 \times 10^{30}}{1.00000 \times 10^8} = 1.00000 \times 10^{22}\ \text{kg/m}$$

$$D_2 = \frac{M_2}{R_2} = \frac{1.00000 \times 10^{24}}{1.00000 \times 10^8} = 1.00000 \times 10^{16}\ \text{kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = \frac{D_1}{D_{crit}} = 1.48523 \times 10^{-5}$$

$$D_{norm,2} = \frac{D_2}{D_{crit}} = 1.48523 \times 10^{-11}$$

Schwarzschild radii:

$$r_{s,1} = 2GM_1/c^2 = 1.48523 \times 10^3\ \text{m}$$

$$r_{s,2} = 2GM_2/c^2 = 1.48523 \times 10^{-3}\ \text{m}$$

Compactness ratio for System 1, computed two ways:

$$k_s = r_{s,1}/R_1 = 1.48523 \times 10^{-5}$$

$$k_d = D_{norm,1} = 1.48523 \times 10^{-5}$$

GTD via r_s/R:

$$\text{GTD}_{s,1} = \sqrt{1 - r_{s,1}/R_1} = 1 - 7.42619 \times 10^{-6}$$

$$\text{GTD}_{s,2} = \sqrt{1 - r_{s,2}/R_2} = 1 - 7.42616 \times 10^{-12}$$

GTD via D/D_{crit}:

$$\text{GTD}_{d,1} = \sqrt{1 - D_{norm,1}} = 1 - 7.42619 \times 10^{-6}$$

$$\text{GTD}_{d,2} = \sqrt{1 - D_{norm,2}} = 1 - 7.42617 \times 10^{-12}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\text{GTD}_{s,1}/\text{GTD}_{s,2} = \frac{1 - 7.42619 \times 10^{-6}}{1 - 7.42616 \times 10^{-12}} = 1 - 7.42618 \times 10^{-6}$$

$$\text{GTD}_{d,1}/\text{GTD}_{d,2} = \frac{1 - 7.42619 \times 10^{-6}}{1 - 7.42617 \times 10^{-12}} = 1 - 7.42619 \times 10^{-6}$$

All values above appear in column $R = R$ of Table 1 in Appendix A.

4.3 Constant Density ($\rho_1 = \rho_2$)

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k:

$$k = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1} = \left(\frac{R_1}{R_2}\right)^2$$

The compactness ratio reduces to the square of the radius ratio. The linear scale factor follows as $\sqrt{k} = R_1/R_2$, recovering the isometric equivalence.

Example (Constant Density): System 1 is the system being measured; System 2 is the comparator at its Schwarzschild radius. S_1: $M_1 = 7.63040 \times 10^{28}$ kg, $R_1 = 1.00000 \times 10^3$ m. S_2: $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3$ m. Both at $\rho = 1.82181 \times 10^{19}$ kg/m^3.

DeGerlia compactness for each system:

$$D_1 = \frac{M_1}{R_1} = \frac{7.63040 \times 10^{28}}{1.00000 \times 10^3} = 7.63040 \times 10^{25}\ \text{kg/m}$$

$$D_2 = \frac{M_2}{R_2} = \frac{2.00000 \times 10^{30}}{2.97046 \times 10^3} = 6.73295 \times 10^{26}\ \text{kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = \frac{D_1}{D_{crit}} = \frac{7.63040 \times 10^{25}}{6.73295 \times 10^{26}} = 1.13329 \times 10^{-1}$$

$$D_{norm,2} = \frac{D_2}{D_{crit}} = \frac{6.73295 \times 10^{26}}{6.73295 \times 10^{26}} = 1.00000$$

DeGerlia compactness ratio:

$$k = \frac{D_1}{D_2} = \frac{D_{norm,1}}{D_{norm,2}} = 1.13329 \times 10^{-1}$$

Schwarzschild ratio for System 1 ($r_{s,1} = 2GM_1/c^2 = 1.13329 \times 10^2$ m):

$$k = \frac{r_{s,1}}{R_1} = \frac{1.13329 \times 10^2}{1.00000 \times 10^3} = 1.13329 \times 10^{-1}$$

GTD via k:

$$\sqrt{1-k} = \sqrt{1 - \frac{D_{norm,1}}{D_{norm,2}}} = 1 - 5.83680 \times 10^{-2}$$

GTD via k:

$$\sqrt{1-k} = \sqrt{1 - 1.13329 \times 10^{-1}} = 1 - 5.83680 \times 10^{-2}$$

This case is tabulated in Table 2, column $\rho = \rho$ of Appendix A.

Example (Constant Density, Table Case 3): Two systems with the same mass density. S_1: $M_1 = 1.00000 \times 10^{30}$ kg, $R_1 = 1.00000 \times 10^8$ m. S_2: $M_2 = 1.00000 \times 10^{27}$ kg, $R_2 = 1.00000 \times 10^7$ m.

DeGerlia compactness:

$$D_1 = M_1/R_1 = 1.00000 \times 10^{22}\ \text{kg/m}$$

$$D_2 = M_2/R_2 = 1.00000 \times 10^{20} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = D_1/D_{crit} = 1.48523 \times 10^{-5}$$

$$D_{norm,2} = D_2/D_{crit} = 1.48523 \times 10^{-7}$$

Schwarzschild radii:

$$r_{s,1} = 2GM_1/c^2 = 1.48523 \times 10^3 \text{ m}$$

$$r_{s,2} = 2GM_2/c^2 = 1.48523 \times 10^0 \text{ m}$$

Schwarzschild ratio for System 1:

$$k = r_{s,1}/R_1 = 1.48523 \times 10^{-5}$$

GTD via r_s/R:

$$\text{GTD}_{s,1} = \sqrt{1 - r_{s,1}/R_1} = 1 - 7.42619 \times 10^{-6}$$

$$\text{GTD}_{s,2} = \sqrt{1 - r_{s,2}/R_2} = 1 - 7.42616 \times 10^{-8}$$

GTD via D/D_{crit}:

$$\text{GTD}_{d,1} = \sqrt{1 - D_{norm,1}} = 1 - 7.42619 \times 10^{-6}$$

$$\text{GTD}_{d,2} = \sqrt{1 - D_{norm,2}} = 1 - 7.42617 \times 10^{-8}$$

All values above appear in column $\rho = \rho$ of Table 1 in Appendix A.

4.4 Constant Inertia ($I_1 = I_2$)

When the two systems share the same moment of inertia, $M_1 R_1^2 = M_2 R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k:

$$k = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1} = \left(\frac{R_2}{R_1}\right)^3$$

The compactness ratio reduces to the cube of the radius ratio.

Example (Constant Inertia): System 1 is the system being measured; System 2 is the comparator at its Schwarzschild radius. S_1: $M_1 = 1.76450 \times 10^{29}$ kg, $R_1 = 1.00000 \times 10^4$ m. S_2: $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3$ m. Both have $I = MR^2 = 1.76450 \times 10^{37}$ kg·m^2.

DeGerlia compactness for each system:

$$D_1 = \frac{M_1}{R_1} = \frac{1.76450 \times 10^{29}}{1.00000 \times 10^4} = 1.76450 \times 10^{25} \text{ kg/m}$$

$$D_2 = \frac{M_2}{R_2} = \frac{2.00000 \times 10^{30}}{2.97046 \times 10^3} = 6.73295 \times 10^{26} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = \frac{D_1}{D_{crit}} = \frac{1.76450 \times 10^{25}}{6.73295 \times 10^{26}} = 2.62069 \times 10^{-2}$$

$$D_{norm,2} = \frac{D_2}{D_{crit}} = \frac{6.73295 \times 10^{26}}{6.73295 \times 10^{26}} = 1.00000$$

DeGerlia compactness ratio:

$$k = \frac{D_1}{D_2} = \frac{D_{norm,1}}{D_{norm,2}} = 2.62069 \times 10^{-2}$$

Schwarzschild ratio for System 1 ($r_{s,1} = 2GM_1/c^2 = 2.62069 \times 10^2$ m):

$$k = \frac{r_{s,1}}{R_1} = \frac{2.62069 \times 10^2}{1.00000 \times 10^4} = 2.62069 \times 10^{-2}$$

GTD via k:

$$\sqrt{1-k} = \sqrt{1 - \frac{D_{norm,1}}{D_{norm,2}}} = 1 - 1.31905 \times 10^{-2}$$

GTD via k:

$$\sqrt{1-k} = \sqrt{1 - 2.62069 \times 10^{-2}} = 1 - 1.31905 \times 10^{-2}$$

This case is tabulated in Table 2, column $I = I$ of Appendix A.

Example (Constant Inertia, Table Case 4): Two systems with the same moment of inertia. S_1: $M_1 = 1.00000 \times 10^{32}$ kg, $R_1 = 1.00000 \times 10^7$ m. S_2: $M_2 = 1.00000 \times 10^{30}$ kg, $R_2 = 1.00000 \times 10^8$ m.

DeGerlia compactness:

$$D_1 = M_1/R_1 = 1.00000 \times 10^{25} \text{ kg/m}$$

$$D_2 = M_2/R_2 = 1.00000 \times 10^{22} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = D_1/D_{crit} = 1.48523 \times 10^{-2}$$

$$D_{norm,2} = D_2/D_{crit} = 1.48523 \times 10^{-5}$$

Schwarzschild radii:

$$r_{s,1} = 2GM_1/c^2 = 1.48523 \times 10^5 \text{ m}$$

$$r_{s,2} = 2GM_2/c^2 = 1.48523 \times 10^3 \text{ m}$$

Schwarzschild ratio for System 1:

$$k = r_{s,1}/R_1 = 1.48523 \times 10^{-2}$$

GTD via r_s/R:

$$\text{GTD}_{s,1} = \sqrt{1 - r_{s,1}/R_1} = 1 - 7.45394 \times 10^{-3}$$

$$\text{GTD}_{s,2} = \sqrt{1 - r_{s,2}/R_2} = 1 - 7.42619 \times 10^{-6}$$

GTD via D/D_{crit}:

$$\text{GTD}_{d,1} = \sqrt{1 - D_{norm,1}} = 1 - 7.45395 \times 10^{-3}$$

$$\text{GTD}_{d,2} = \sqrt{1 - D_{norm,2}} = 1 - 7.42619 \times 10^{-6}$$

All values above appear in column $I = I$ of Table 1 in Appendix A.

4.5 General Case (unconstrained)

When no constraint is applied between the systems, k stays in its full form:

$$k = \frac{M_1 R_2}{M_2 R_1}$$

Both mass and radius must be specified independently for each system.

Example (General Case): System 1 is the system being measured; System 2 is the comparator at its Schwarzschild radius. S_1: $M_1 = 1.00000 \times 10^{29}$ kg, $R_1 = 1.00000 \times 10^4$ m. S_2: $M_2 = 2.00000 \times 10^{30}$ kg, $R_2 = r_s = 2GM_2/c^2 = 2.97046 \times 10^3$ m.

DeGerlia compactness ratio:

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{(1.00000 \times 10^{29})(2.97046 \times 10^3)}{(2.00000 \times 10^{30})(1.00000 \times 10^4)} = 1.48523 \times 10^{-2}$$

Schwarzschild ratio for System 1 ($r_{s,1} = 2GM_1/c^2 = 1.48523 \times 10^2$ m):

$$k = \frac{r_{s,1}}{R_1} = \frac{1.48523 \times 10^2}{1.00000 \times 10^4} = 1.48523 \times 10^{-2}$$

GTD via k:

$$\sqrt{1-k} = \sqrt{1 - 1.48523 \times 10^{-2}} = 1 - 7.45394 \times 10^{-3}$$

GTD via k:

$$\sqrt{1-k} = \sqrt{1 - 1.48523 \times 10^{-2}} = 1 - 7.45394 \times 10^{-3}$$

This case is tabulated in Table 2, column General of Appendix A.

Example (General Case, Table General): An unconstrained pair. S_1: $M_1 = 1.00000 \times 10^{20}$ kg, $R_1 = 2.00000 \times 10^{-7}$ m. S_2: $M_2 = 1.00000 \times 10^{30}$ kg, $R_2 = 1.00000 \times 10^7$ m.

DeGerlia compactness:

$$D_1 = M_1/R_1 = 5.00000 \times 10^{26} \text{ kg/m}$$

$$D_2 = M_2/R_2 = 1.00000 \times 10^{23} \text{ kg/m}$$

Normalized to the DeGerlia Threshold:

$$D_{norm,1} = D_1/D_{crit} = 7.42617 \times 10^{-1}$$

$$D_{norm,2} = D_2/D_{crit} = 1.48523 \times 10^{-4}$$

Schwarzschild radii:

$$r_{s,1} = 2GM_1/c^2 = 1.48523 \times 10^{-7} \text{ m}$$

$$r_{s,2} = 2GM_2/c^2 = 1.48523 \times 10^3 \text{ m}$$

Schwarzschild ratio for System 1:

$$k = r_{s,1}/R_1 = 7.42616 \times 10^{-1}$$

GTD via r_s/R:

$$\text{GTD}_{s,1} = \sqrt{1 - r_{s,1}/R_1} = 1 - 4.92670 \times 10^{-1}$$

$$\text{GTD}_{s,2} = \sqrt{1 - r_{s,2}/R_2} = 1 - 7.42644 \times 10^{-5}$$

GTD via D/D_{crit}:

$$\text{GTD}_{d,1} = \sqrt{1 - D_{norm,1}} = 1 - 4.92670 \times 10^{-1}$$

$$\text{GTD}_{d,2} = \sqrt{1 - D_{norm,2}} = 1 - 7.42644 \times 10^{-5}$$

All values above appear in column General of Table 1 in Appendix A.

4.6 Classical Static Density Example

Setup. A uniform sphere of mass 1 kg and radius 1 m (System 2) is scaled isometrically by a factor of two, producing a sphere of mass $2^3 = 8$ kg and radius 2 m (System 1). The linear scale factor between them is $k = R_1/R_2 = 2$ by construction.

Change in inertia. The moments of inertia are

$$I_1 = \tfrac{2}{5}(8)(2)^2 = 12.8\,\text{kg m}^2, \qquad I_2 = \tfrac{2}{5}(1)(1)^2 = 0.4\,\text{kg m}^2,$$

so $I_1/I_2 = 32 = 2^5 = k^5$ — the moment of inertia changes by the fifth power of the linear scale factor.

Gravitational time dilation. Each system's Schwarzschild radius is $r_s = 2GM/c^2$, giving $r_{s,1} = 1.18819 \times 10^{-26}$ m and $r_{s,2} = 1.48523 \times 10^{-27}$ m. The Schwarzschild gravitational time dilation $gtd = \sqrt{1 - r_s/R}$ is then

$$gtd_1 = \sqrt{1 - \frac{r_{s,1}}{R_1}} = 1 - 2.97046 \times 10^{-27},$$

$$gtd_2 = \sqrt{1 - \frac{r_{s,2}}{R_2}} = 1 - 7.42616 \times 10^{-28},$$

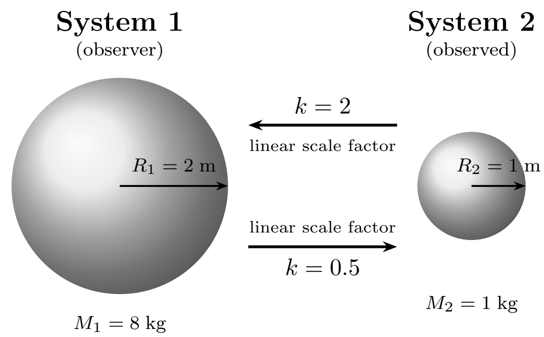
so the relative time dilation between the two systems is

$$\frac{gtd_1}{gtd_2} = 1 - 2.22785 \times 10^{-27}.$$

DeGerlia compactness ratio. The ratio k follows from the DeGerlia compactness $D = M/R$ ($D_1 = 4$ kg/m, $D_2 = 1$ kg/m):

$$k = \frac{D_1}{D_2} = \frac{4}{1} = 4 = k^2, \qquad \sqrt{k} = 2 = k.$$

This matches the Classical Static Density Example column of Appendix A: the gravitational time dilation ratio is $1 - 2.22785 \times 10^{-27}$ and $\sqrt{k} = 2$.



Two uniform spheres related by isometric scaling, scale factor $k = 2$: System 1 ($M_1 = 8\,kg$, $R_1 = 2\,m$) and System 2 ($M_2 = 1\,kg$, $R_2 = 1\,m$). The moment of inertia changes by $2^5 = 32$; the relative time dilation $1 - 2.22785 \times 10^{-27}$ is obtained equivalently from the Schwarzschild and inertial-density formulations.

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity-deriving-gr-sr-from-ste`.

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Appendix A Pair Comparison Tables

Tables 1 and 2 present the complete system properties and derived ratios for six system pairs spanning the constraint cases from the derivations in Section 4: static mass (M=M), static radius (R=R), constant density ($\rho = \rho$), constant inertia (I=I), an unconstrained general case, and an STE example. The three algebraic forms of k in Equation 4 produce identical values in every case, alongside the gravitational time-dilation ratio computed independently from the standard Schwarzschild form.

Table 1 System properties for each pair comparison case.

	Case 1 ($M = M$)	Case 2 ($R = R$)	Case 3 ($\rho = \rho$)	Case 4 ($I = I$)	General	Classical ($\rho = \rho$)
m_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+32	1.00000e+20	8.00000e+0
m_2 (kg)	2.00000e+30	1.00000e+24	1.00000e+27	1.00000e+30	1.00000e+30	1.00000e+0
r_1 (m)	3.00000e+3	1.00000e+8	1.00000e+8	1.00000e+7	2.00000e-7	2.00000e+0
r_2 (m)	7.00000e+8	1.00000e+8	1.00000e+7	1.00000e+8	1.00000e+7	1.00000e+0
ρ_1 (kg/m^3)	1.76839e+19	2.38732e+5	2.38732e+5	2.38732e+10	2.98416e+39	2.38732e-1
ρ_2 (kg/m^3)	1.39203e+3	2.38732e-1	2.38732e+5	2.38732e+5	2.38732e+8	2.38732e-1
I_1 (kg·m^2)	7.20000e+36	4.00000e+45	4.00000e+45	4.00000e+45	1.60000e+6	1.28000e+1
I_2 (kg·m^2)	3.92000e+47	4.00000e+39	4.00000e+40	4.00000e+45	4.00000e+43	4.00000e-1
D_1 (kg/m)	6.66667e+26	1.00000e+22	1.00000e+22	1.00000e+25	5.00000e+26	4.00000e+0
D_2 (kg/m)	2.85714e+21	1.00000e+16	1.00000e+20	1.00000e+22	1.00000e+23	1.00000e+0
r_{s_1} (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+5	1.48523e-7	1.18819e-26
r_{s_2} (m)	2.97046e+3	1.48523e-3	1.48523e+0	1.48523e+3	1.48523e+3	1.48523e-27
$D1_{\_norm}$	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
$D2_{\_norm}$	4.24352e-6	1.48523e-11	1.48523e-7	1.48523e-5	1.48523e-4	1.48523e-27
k_s	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42616e-1	5.94093e-27
k_d	9.90155e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
GTD$_{s,1}$	1-9.00777e-1	1-7.42619e-6	1-7.42619e-6	1-7.45394e-3	1-4.92670e-1	1-2.97046e-27
GTD$_{s,2}$	1-2.12176e-6	1-7.42616e-12	1-7.42616e-8	1-7.42619e-6	1-7.42644e-5	1-7.42616e-28
GTD$_{d,1}$	1-9.00780e-1	1-7.42619e-6	1-7.42619e-6	1-7.45395e-3	1-4.92670e-1	1-2.97047e-27
GTD$_{d,2}$	1-2.12176e-6	1-7.42617e-12	1-7.42617e-8	1-7.42619e-6	1-7.42644e-5	1-7.42617e-28

Table 2 Derived ratios across constraint cases.

	Case 1 ($M = M$)	Case 2 ($R = R$)	Case 3 ($\rho = \rho$)	Case 4 ($I = I$)	General	Classical ($\rho = \rho$)
$\sqrt{D_1/D_2}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(m_1 r_2)/(m_2 r_1)}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(I_1/I_2)(r_2/r_1)^3}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.83046e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(m_1 r_2)/(m_2 r_1)$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
D_1/D_2	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$(I_1/I_2)(r_2/r_1)^3$	2.33333e+5	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
GTD$_{s,1}$/GTD$_{s,2}$	1-9.00776e-1	1-7.42618e-6	1-7.35193e-6	1-7.44657e-3	1-4.92632e-1	1-2.22785e-27
GTD$_{d,1}$/GTD$_{d,2}$	1-9.00780e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
I_1/I_2	1.83673e-11	1.00000e+6	1.00000e+5	1.00000e+0	4.00000e-38	3.20000e+1
$(I_1/I_2)^{1/2}$	4.28571e-6	1.00000e+3	3.16228e+2	1.00000e+0	2.00000e-19	5.65685e+0
$(\rho_1/\rho_2)^{1/5}(r_1/r_2)$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
$(I_1/I_2)^{1/5}$	7.12540e-3	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
m_1/m_2	1.00000e+0	1.00000e+6	1.00000e+3	1.00000e+2	1.00000e-10	8.00000e+0
$\sqrt{m_1/m_2}$	1.00000e+0	1.00000e+3	3.16228e+1	1.00000e+1	1.00000e-5	2.82843e+0
r_1/r_2	4.28571e-6	1.00000e+0	1.00000e+1	1.00000e-1	2.00000e-14	2.00000e+0
$\sqrt{r_1/r_2}$	2.07020e-3	1.00000e+0	3.16228e+0	3.16228e-1	1.41421e-7	1.41421e+0